regardless of how we rotate it around in our hands or how we shift the angle from which we view it, we say that the universe exemplifies strong force symmetry: Physics is unchanged by—it is completely insensitive to—these force-charge shifts. For historical reasons, physicists also say that the strong force symmetry is an example of a gauge symmetry.

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Here is the essential point. Just as the symmetry between all possible observational vantage points in general relativity requires the existence of the gravitational force, developments relying on work of Hermann Weyl in the 1920s and Chen-Ning Yang and Robert Mills in the 1950s showed that gauge symmetries require the existence of yet other forces. Much like a sensitive environmental-control system that keeps temperature, air pressure, and humidity in an area completely constant by compensating perfectly for any exterior influences, certain kinds of force fields, according to Yang and Mills, will provide perfect compensation for shifts in force charges, thereby keeping the physical interactions between the particles completely unchanged. For the case of the gauge symmetry associated with shifting quark-color charges, the required force is none other than the strong force itself. That is, without the strong force, physics would change under the kinds of shifts of color charges indicated above. This realization shows that, although the gravitational force and the strong force have vastly different properties (recall, for example, that gravity is far feebler than the strong force and operates over enormously larger distances), they do have a somewhat similar heritage: they are each required in order that the universe embody particular symmetries. Moreover, a similar discussion applies to the weak and electromagnetic forces, showing that their existence, too, is bound up with yet other gauge symmetries —the so-called weak and electromagnetic gauge symmetries. And hence, all four forces are directly associated with principles of symmetry.

This common feature of the four forces would seem to bode well for the suggestion made at the beginning of this section. Namely, in our effort to incorporate quantum mechanics into general relativity we should seek a quantum field theory of the gravitational force, much as physicists have discovered successful quantum field theories of the